Roadmap. PDF pp. 1–2 (§12): mass dimensions; the sign of $[g]$ sets high-energy behavior and, by power counting, renormalizability (LecNote14, §18). PDF pp. 3–7 (§13), with no perturbation theory: the exact (hatched) propagator is a pole at the physical mass with residue 1 plus a continuum from $s = 4m^2$. LecNote12 (§§14–15) imposes this at one loop: $\Pi(-m^2) = \Pi'(-m^2) = 0$. Supplements: AsymptoticStates, FeynmanPropagator, SpectralRepresentation, DimensionalAnalysis_hbar.

See. 12. Dimensional Analysis.

Naïve.

$\hbar = c = 1.$

$\uparrow \quad \uparrow$

$E \cdot \not{\Sigma} T \qquad L/T.$

$\hbar c = E \cdot L.$

$= 0.2\ \text{GeV} \cdot \text{eV}.$

So we can convert anything to mass (energy) dimension.

$[m] = +1$

$[x^\mu] = -1, \quad [dx] = -1.$

$[\partial^\mu] = +1$

$[d^d x] = \not{d}. - d.$

for a general theory

$$\mathcal{L} = -\frac{1}{2}\partial^\mu\varphi\,\partial_\mu\varphi - \frac{1}{2}m^2\varphi^2 - \sum_{n=3}^{N}\frac{1}{n!}g_n\varphi^n.$$ (1)

$$S = \int d^d x\,\mathcal{L}.$$

$$Z(J) = \int \mathcal{D}\varphi\, e^{\frac{i}{\hbar}\int d^d x\,(\mathcal{L} + J\varphi)}$$

1 The sign of $[g_n]$ (PDF p. 2) sorts couplings: $[g] > 0$ super-renormalizable, $[g] = 0$ renormalizable, $[g] < 0$ non-renormalizable (LecNote14, §18). $[g_3] = 0$ at $d = 6$, hence Srednicki's φ^3 in six dimensions. §12 also keeps $\hbar$: internal propagators carry $\hbar$, vertices $1/\hbar$, so an L-loop diagram goes as $\hbar^{L-1}$. Loops are an expansion in $\hbar$.

because $[S]=0$ as it is the ~~[illegible]~~ exponent.

$[L] = d.$

$[\varphi] = \frac{1}{2}(d-2).$

$[\bar{g}_n] = d - \frac{1}{2}n(d-2).$

$[g_3] = \frac{1}{2}(6-d).$

One concept is the non-trivial dependence ~~not~~ on couplings for scattering amplitudes. need to be dimensionless argument of a function.

For high energy, it means

$f(\bar{g} s^{-[g]/2}).$ $\quad \bar{g} = \left(\frac{g}{\Lambda}\right)^{[g]}$ ①

So for $[g] < 0$, $f(\bar{g} s^{-[g]/2})$ has a blowing up requirement.

So for $[g] > 0$, $f(\bar{g} s^{-[g]/2})$ becomes trivial at high energy.

for $[g] = 0$, $f(g)$ will be fine across scales.

1 Should read $f(g\, s^{-[g]/2})$: $[s] = 2$, so this is already dimensionless. With a reference scale, $\bar{g} \equiv g/\Lambda^{[g]}$ and the argument is $\bar{g}\,(\Lambda^2/s)^{[g]/2}$. As written, $(g/\Lambda)^{[g]}$ has dimension $[g]([g]-1)$.

Sec 13: Propagator & Corrections

Before begin the class, ask for feedback for the first part of the course.

(Basic of QFT, scalar theories, Feynman Rules, etc.)

Emphasize QFT is to be digested all the time.

We've defined the "exact" propagator via

$$\Delta(x-y) \equiv -i\langle 0|T(\varphi(x)\varphi(y)|0\rangle$$

with

$$\langle 0|\varphi(x)|0\rangle = 0, \quad \langle k|\varphi(x)|0\rangle = e^{-ikx}$$

In d-spacetime dimensions

$$\langle k|k'\rangle = (2\pi)^{d-1} 2\omega \delta^{d-1}(\vec{k}-\vec{k}').$$

[here I assumed only 1 time direction $SO^+(1,d-1)$]

where $\omega = \sqrt{|\vec{k}|^2+m^2}$

the corresponding completeness relation yields

$$\int \widetilde{dk}\, |k\rangle\langle k| = \underset{\uparrow}{I_1}$$

identity op in one-particle subspace.

$$\widetilde{dk} \equiv \frac{d^{d-1}\vec{k}}{(2\pi)^{d-1}\, 2\omega_{\vec{k}}}$$

is the Lorentz invariant phase-space differential

We also define the exact momentum space propagator

$\hat{\Delta}(k^2)$ via

$$\Delta(x-y) = \int \frac{d^d k}{(2\pi)^d} e^{-ik(x-y)} \hat{\Delta}(k^2)$$

In free field theory, the momentum-space propagator is

$$\hat{\Delta}(k^2) = \frac{1}{k^2 + m^2 - i\epsilon}$$

It has an isolated pole at $k^2 = -m^2$ with residue one

m is the actual, physical mass of the free particle that enters into the energy momentum relation.

Let's insert a complete set of energy eigenstates.

(including $|0\rangle$, $|k\rangle$, and multiparticle state $|k, n\rangle$ ← total momentum with mass M.

Then (Note that I haven't used perturbative expansion in the following)

$$\langle 0|\varphi(x)\varphi(y)|0\rangle$$

$$= \langle 0|\varphi(x)|0\rangle\langle 0|\varphi(y)|0\rangle + \int \hat{dk} \langle 0|\varphi(x)|k\rangle\langle k|\varphi(y)|0\rangle$$

$$+ \sum_n \int \hat{dk} \langle 0|\varphi(x)|k,n\rangle\langle k,n|\varphi(y)|0\rangle$$

Recall that we previously showed (not fully but pretty good) $\langle k,n|k\rangle = 0$.

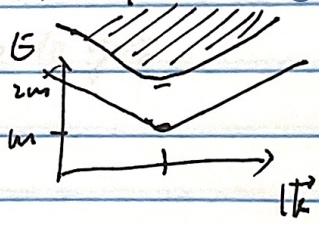
$\int dk$ would also work.

mean integrating over all phase space.

Noting that $\varphi(x) = e^{-i\hat{P}^\mu x_\mu}\varphi(0)e^{i\hat{P}^\mu x_\mu}$

$$\langle k,n|\varphi(x)|0\rangle = e^{-ikx}\langle k,n|\varphi(0)|0\rangle \quad (1)$$

where $k^0 = (\vec{k}^2 + M^2)^{1/2}$



1 Constant modulus: this never decays. What vanished as $t \to \pm\infty$ (red box) is the packet-smeared $\langle k,n|a_1^\dagger(t)|0\rangle$ (AsymptoticStates). These states stay, in $\rho(s)$, but leave the pole untouched: $\rho = 0$ below $4m^2$.

$$\langle 0|\varphi(x)\varphi(y)|0\rangle = \int \hat{dk}\, e^{-ik(x-y)} + \sum_n \int \hat{dk}\, e^{-ik(x-y)} |\langle k,n|\varphi(0)|0\rangle|^2.$$

We can define a quantity called spectral density

$$\rho(s) \equiv \sum_n |\langle k,n|\varphi(0)|0\rangle|^2 \delta(s - M^2)$$

Recall earlier we showed that $\lim_{t\to\pm\infty} \langle k,n|\varphi(x)|0\rangle \to 0$. Here is not that limit. And k goes "off-shell" from 1 particle state.

$\rho(s) \geq 0$ for $s \geq 4m^2$ and $\rho(s) = 0$ for $s < 4m^2$

↓

preserves unitarity

ensures microcausality (non-negative probability)

1 Insert $1 = \int ds\, \delta(s - M^2)$ under $\sum_n$. ρ is Lorentz invariant ($\varphi(0)$ scalar, $|0\rangle$ invariant), so it depends on k only via $k^2 = -s$ and leaves $\int \widetilde{dk}$; it vanishes below $4m^2$, two particles at rest (barring bound states). Each s-term is a free two-point function of mass^2 s: the identity below applies with $m^2 \to s$.

Now we get

$$\langle 0|\varphi(x)\varphi(y)|0\rangle = \int \widetilde{dk}\, e^{-ik(x-y)} + \int_{4m^2}^{\infty} ds\, \rho(s) \int \widetilde{dk}\, e^{-ik(x-y)} \quad (1)$$

here $k^0 = (\vec{k}^2 + m^2)^{1/2}$

$k^0 = (\vec{k}^2 + s)^{1/2}$

We can swap x and y to get

$$\langle 0|\varphi(y)\varphi(x)|0\rangle = \int \widetilde{dk}\, e^{-ik(x-y)} + \int_{4m^2}^{\infty} ds\, \rho(s) \int \widetilde{dk}\, e^{-ik(x-y)}$$

We can combine the ~~above~~ above two to get time-ordered product.

$$\langle 0|T\varphi(x)\varphi(y)|0\rangle = \theta(x^0 - y^0)\langle 0|\varphi(x)\varphi(y)|0\rangle + \theta(y^0 - x^0)\langle 0|\varphi(y)\varphi(x)|0\rangle$$

We're already seen a useful math in early lecture on path integral for free theory, that.

$$\int \frac{d^d k}{(2\pi)^d} \frac{e^{ik(x-y)}}{k^2 + m^2 - i\epsilon} = i\theta(x^0 - y^0) \int \widetilde{dk}\, e^{ik(x-y)} + i\theta(y^0 - x^0) \int \widetilde{dk}\, e^{-ik(x-y)}$$

So we get

$$i\langle 0|T\varphi(x)\varphi(y)|0\rangle = \int \frac{d^dk}{(2\pi)^d} e^{-ik(x-y)} \left[\frac{1}{k^2+m^2-i\epsilon} + \int_{4m^2}^{\infty} ds\, \rho(s) \frac{1}{k^2+s-i\epsilon}\right]$$ (1)

1 Two more consequences of $\rho \geq 0$. The equal-time commutator $[\varphi(\mathbf{x}), Z_\varphi \dot{\varphi}(\mathbf{y})] = i\delta^{d-1}(\mathbf{x}-\mathbf{y})$ gives $1 + \int ds\, \rho(s) = 1/Z_\varphi$, so $0 \leq Z_\varphi \leq 1$. For $k^2 > 0$ every term is positive: $\tilde{\Delta} \geq 1/(k^2+m^2)$, so no propagator falls faster than $1/k^2$. Pauli–Villars (LecNote12) breaks this: its regulator is a state with $\rho < 0$.

2 Should read $\int_{4m^2}^{\infty} ds\, \rho(s)/(k^2+s-i\epsilon)$, as in the "So we get" bracket; s is already a mass squared.

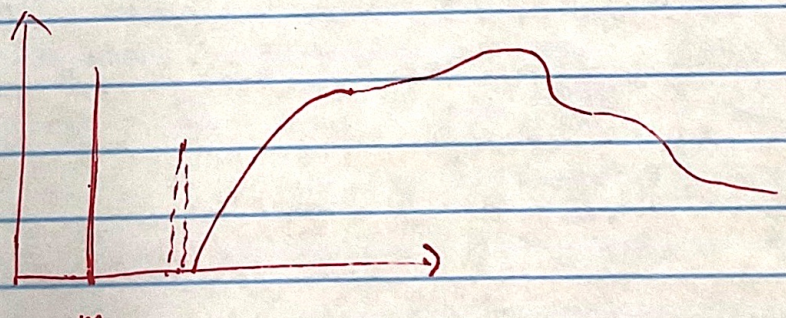
Hence

$$\tilde{\Delta}(k^2) = \frac{1}{k^2+m^2-i\epsilon} + \int_{4m^2}^{\infty} ds\, \rho(s) \frac{1}{k^2+s^2-i\epsilon}$$ (2)

Lehmann–Källén form of the exact propagator.

[Note this is actually also true for ~~non-perturbative~~ strongly coupled systems]

free theory pole. (3)



Next, we explore the meaning & definition of m.

3 Only the form is free. This is the exact one-particle pole, at the physical mass m, with residue $|\langle k|\varphi(0)|0\rangle|^2 = 1$ (PDF p. 3); it is the spike in the sketch, so the full weight is $\delta(s-m^2) + \rho(s)$. LecNote12 keeps it there order by order, $\Pi(-m^2) = \Pi'(-m^2) = 0$, which fixes A and B.